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I am writing to apply for the Staff Data Analyst I position in the University of Pittsburgh Department of Neurology. The role is a direct fit for how I work: taking data-rich, ambiguous problems, building rigorous statistical models, and presenting the findings in terms that make sense to clinicians and non-statisticians alike. Real-world evidence work for seizure diagnostics and clinical trials is exactly the kind of consequential analysis I want my career to be built around, and the explicit training pathway in EHR methods, clinical trial methodology, and applied AI tells me this is a team that invests in its analysts.

The strongest proof I can offer is a project that ran from a loosely defined brief to a quantified outcome for a client with no tolerance for hedged analysis. As the sole analyst on a restaurant chain operating across Australia and New Zealand, I structured an ambiguous cost-overspend concern into a testable statistical question, built pricing-movement and cost-leakage models, and presented the findings directly to the CFO, CPO, and category managers in plain, non-technical terms. The result was approximately A\$2M per year in identified savings in their largest procurement category. The skill that mattered most was not the modelling itself, it was framing the question carefully, communicating the uncertainty honestly, and making the output actionable for people who had not run the analysis. That is precisely what the Neurology group needs from someone helping researchers understand what their seizure treatment data is and is not showing.

On the technical side, my statistical foundations are strong and applied: multivariable regression, binary classification, time-series forecasting, anomaly detection, cross-validation, feature engineering, all built in Python on production datasets. For the EHR-specific techniques the role trains on, including propensity matching, survival models, and GLM variants beyond logistic regression, I am genuinely excited to ramp under clinical guidance. On the AI and LLM side I am already well ahead of the training curve: I have built multi-agent systems in production, applied rigorous AI-output evaluation in client-facing workflows, and won a company hackathon with an LLM pipeline that was later adopted company-wide. Where the JD says it will train on ML/AI and LLMs, I am bringing an existing foundation that clinical context will only deepen. On R specifically: I am Python-native and actively building R fluency. My statistical foundations transfer cleanly; I am committed to a focused sprint on tidyverse and ggplot2 before starting.

The University of Pittsburgh Department of Neurology is doing the kind of work I want to contribute to, combining real-world evidence with clinical trial analysis to improve outcomes for people with seizure disorders. My background is in applied analytics across complex, large-scale datasets, with a consistent emphasis on data quality, careful handling of sensitive information, and communicating quantitative results to non-technical audiences. The domain is new to me; the analytical instincts, the rigor, and the communication skills are already here. I would welcome the chance to discuss how my background maps to the work the group is doing.

Sincerely,

Dhruv Nirmal